

# Fan Zhang

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## Summary

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Geospatial data scientist with a Ph.D. in Environmental Geoscience and 5+ years' experience applying satellite-based analytics to environmental forecasting and pattern/anomaly detection. Specialized in NDVI, EVI, and TVI trend analysis, rainfall deviation mapping, vegetation stress modelling, and time-series correlation for yield prediction. Skilled in building automated pipelines that integrate multi-sensor data (Sentinel-1/2, Landsat 8/9, MODIS, Radarsat-2, RCM) with Python-based machine learning and statistical models to generate actionable insights.

## Core Skills

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- **Geospatial Forecasting and Anomaly Detection:** ArcGIS, QGIS, Google Earth Engine, geopandas, netCDF, ENVI, PCI Geomatics, esa\_snappy, GMTSAR
- **Programming and Modelling:** Python (pandas, Numpy, xarray), R, MATLAB, JavaScript
- **Machine Learning Tools:** scikit-learn, TensorFlow, PyTorch
- **Geospatial Data Sources:** Radarsat-2, RCM, Sentinel-1, Sentinel-2, Landsat, MODIS, Copernicus ECMWF datasets
- **Professional Certifications:** Google Data Analytics certification, Machine Learning Scientist with Python
- **Languages:** English (fluent/advanced/professional), Chinese (native-speaking)
- **Personality:** Self-motivated; big-picture perspective; able to work collaboratively and independently

## Work Experience

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**Remote Sensing Specialist, WSP E&I (Ottawa, Canada)** Nov 2023 — present

- Developed and deployed machine learning classification and regression models for large-scale geospatial datasets, including vegetation indices and environmental variables.
- Automated ingestion, preprocessing, and model training pipelines in Python.
- Integrated cloud-based workflows (Google Earth Engine) for multi-temporal NDVI and rainfall anomaly monitoring.
- Delivered predictive mapping products to support environmental and resource management decisions.

## **Climate Model Specialist, Geosapiens (Canada)**

Aug 2022 — Aug 2023

- Collected, processed, and analyzed geospatial datasets from various climate change scenarios, including precipitation and temperature anomalies.
- Developed visualization tools for model calibration and verification using NDVI-like vegetation indices and climate variables.
- Estimated environmental variables from reanalysis datasets (ERA5) for statistical models integration.
- Conducted time-series statistical analyses to identify correlations between environmental factors and modelled outcomes.

## **Education**

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**Global Institute for Water Security, University of Saskatchewan, Canada**

Doctor of Philosophy

## **Selected Projects**

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**Coastal Marine Habitat and Vegetation Monitoring (2024)** Client: Fisheries and Oceans Canada

- Integrated airborne LiDAR point cloud data with Sentinel-2 multispectral imagery to conduct coastal habitat and vegetation classification
- Computed vegetation structure metrics (NDVI and canopy height model) to monitor spatial and temporal changes in coastal marine vegetation.
- Applied machine learning classification to differentiate vegetation types and assess habitat extent, achieving 89% classification accuracy
- Automated geospatial workflows in ArcGIS and Google Earth Engine, enabling scalable monitoring for coastal ecosystem health.

## **Chlorophyll-a analysis - Outarde and Manicouagan estuary (2023)**

Client: Hydro Quebec

- Conduct chlorophyll-a time-series **data cleaning** and **analyzing**
- Automated preprocessing and feature engineering workflows in cloud computing platform (**Google Earth Engine**) and Python
- Use the cleaned data to train and test the model to estimate the chlorophyll-a concentration in the St. Lawrence River, improved the model accuracy by 33%

**Lakeside Agricultural Health Monitoring, Xingkai Lake, China (2021)** Client: In-house research

- Monitored crop health and vegetation dynamics in lakeside agricultural zones using multi-temporal Sentinel-2 imagery
- Calculated NDVI, EVI, and other vegetation indices to assess crop vigorous and detect spatial anomalies factors.
- Automated data preprocessing, index calculation, and anomaly detection workflows in Python for repeatable seasonal monitoring.